

1. Linux

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Linux is the backbone of most data engineering tools and systems. Whether you're running Apache Spark on a cluster or scheduling jobs with Airflow, you'll likely do it on a **Linux-based system**.

1.1 Why Linux for Data Engineering?

Reason	Explanation
Open-source & Free	No license required, customizable for different use cases
Command-line Power	Automate tasks and manage servers efficiently
Tool Compatibility	Most big data tools (Hadoop, Spark, Kafka) run on Linux
Security & Permissions	Fine-grained control over access and execution rights
Server Deployment	Most cloud servers (AWS EC2, GCP VM, Azure VMs) run on Linux

1.2 Linux File System Structure

Everything in Linux is a **file** (even devices and processes).

Directory	Purpose
/	Root directory (everything starts here)
/home/	User directories (like /home/krishna)
/bin/	Essential command binaries (ls, cp)
/etc/	Configuration files
/var/	Log files, spool files
/tmp/	Temporary files
/usr/	Applications and utilities

1.3 Terminal

The **terminal** is a command-line interface (CLI) where you run commands.

Popular shells:

- bash (most common)
- zsh
- Fish

What is the difference between Terminal, Shell, and Bash?

Term	What it means	Example
Terminal	The window or program where you type commands (interface)	Ubuntu Terminal, Mac Terminal
Shell	The program that interprets your commands and runs them	Bash, Zsh, Fish
Bash	A popular type of shell (Bourne Again Shell)	bash (default on most Linux systems)

Terminal is the door, Shell is the brain inside, Bash is one popular brain.

1.4 Permissions in Linux

Linux has a powerful permission system. Each file has:

- **Owner**
- **Group**
- **Others**

Permissions are shown like:

-rwxr-xr--

Symbol	Meaning
r	Read
w	Write
x	Execute

To change permissions:

```
chmod +x script.sh      # Makes script executable
```

To change ownership:

```
chown <username>:users file.txt
```

1.4 Linux Role

Linux is **mission-critical** for data engineers because:

Task	Linux Role
Running Spark, Kafka, Hadoop	These run on Linux-based systems
Automating ETL jobs	Bash or Python scripts via cron or Airflow
Managing cloud VMs	SSH into Linux-based EC2/GCP instances
Monitoring and logging	Use top, htop, tail, grep, systemctl etc.
Working with data files	Use awk, sed, cut, sort, cat
Containerization (Docker)	Typically runs on Linux kernels

1.5 Popular Linux Distros

Distro	Description
Ubuntu	Beginner-friendly, widely used
CentOS	Stable, server-focused (RedHat-based)
Debian	Foundation for Ubuntu, stable
Alpine	Lightweight, used in Docker containers

1.6 What are Tar Files?

Tar = Tape Archive

Combines multiple files/folders into one archive.

Often used for backups, data transfer, compression.

Create a tar file:

```
tar -cvf archive.tar folder/
```

Extract a tar file:

```
tar -xvf archive.tar
```

With compression (.tar.gz):

```
tar -czvf archive.tar.gz folder/
```

```
tar -xzvf archive.tar.gz
```

1.7 What are Environment Variables?

- Variables that store system-wide or user-wide settings.

Example:

```
echo $HOME
```

```
# Output: /home/<username>
```

Create your own:

```
export MY_VAR="DataEngineering"
```

```
echo $MY_VAR
```

These are often used to configure tools (like setting database credentials, API keys, paths, etc.)

Make permanent:

- Add to ~/.bashrc or ~/.bash_profile
- ```
export PATH=$PATH:/new/path/here
```

## 1.8 Basic Linux Commands

### Navigating the File System

| Command | What It Does            | Example                   |
|---------|-------------------------|---------------------------|
| pwd     | Print working directory | pwd                       |
| ls      | List files              | ls or ls -l (long format) |
| cd      | Change directory        | cd /home/krishna          |
| cd ..   | Move one level up       | cd ..                     |
| cd ~    | Go to home directory    | cd ~                      |
| mkdir   | Make new directory      | mkdir data                |
| rmdir   | Remove empty directory  | rmdir old_folder          |

### Working with Files

| Command     | What It Does                   | Example                |
|-------------|--------------------------------|------------------------|
| touch       | Create empty file              | touch file.txt         |
| rm          | Delete file                    | rm file.txt            |
| cp          | Copy file                      | cp file1.txt file2.txt |
| mv          | Move/rename file               | mv file.txt archive/   |
| cat         | Display file contents          | cat file.txt           |
| less / more | View file one screen at a time | less file.txt          |
| head / tail | View top/bottom lines of file  | head -n 10 file.txt    |
| nano / vim  | Text editors (inside terminal) | nano file.txt          |

### Permissions and Ownership

| Command | What It Does            | Example                  |
|---------|-------------------------|--------------------------|
| chmod   | Change file permissions | chmod 755 script.sh      |
| chown   | Change file ownership   | chown user:user file.txt |
| ls -l   | View permissions        | ls -l file.txt           |

#### Permissions Breakdown:

-rwxr-xr--

r = read, w = write, x = execute.

### Process & System Management

| Command    | What It Does                  | Example         |
|------------|-------------------------------|-----------------|
| ps         | Show running processes        | ps aux          |
| top / htop | Monitor live system resources | top             |
| kill       | Stop a process                | kill 1234 (PID) |
| df -h      | Show disk usage               | df -h           |
| free -m    | Show memory usage             | free -m         |
| uptime     | Show system uptime            | uptime          |
| whoami     | Display current username      | whoami          |

### Networking Commands

| Command | What It Does                           | Example                                                    |
|---------|----------------------------------------|------------------------------------------------------------|
| ping    | Test network connection                | ping google.com                                            |
| curl    | Fetch URL content (important for APIs) | curl <a href="https://example.com">https://example.com</a> |
| wget    | Download file from internet            | wget <a href="http://file.com">http://file.com</a>         |
| scp     | Secure copy (remote file transfer)     | scp file user@ip:/path/                                    |

### Redirection, Piping & Combining Commands

| Command | Purpose                             | Example                        |
|---------|-------------------------------------|--------------------------------|
| >       | Redirect output to file (overwrite) | echo "hello" > file.txt        |
| >>      | Append output to file               | echo "world" >> file.txt       |
| `       | `                                   | Pipe output to another command |
| ;       | Run multiple commands sequentially  | pwd; ls; whoami                |
| &&      | Run next only if previous succeeds  | mkdir data && cd data          |

#### Superuser / Admin Commands

| Command  | Purpose                          | Example                  |
|----------|----------------------------------|--------------------------|
| sudo     | Run command as admin             | sudo apt update          |
| apt      | Install software (Debian/Ubuntu) | sudo apt install python3 |
| reboot   | Restart system                   | sudo reboot              |
| shutdown | Shut down system                 | sudo shutdown now        |

[Linux Crash Course](#)